Decision-making guide

IT ManagementMicrosimulation

Table of Contents

1. Simulation platform introduction

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes Introduction, Market, System Architecture, Software Development, Security, Innovation, Decision Checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

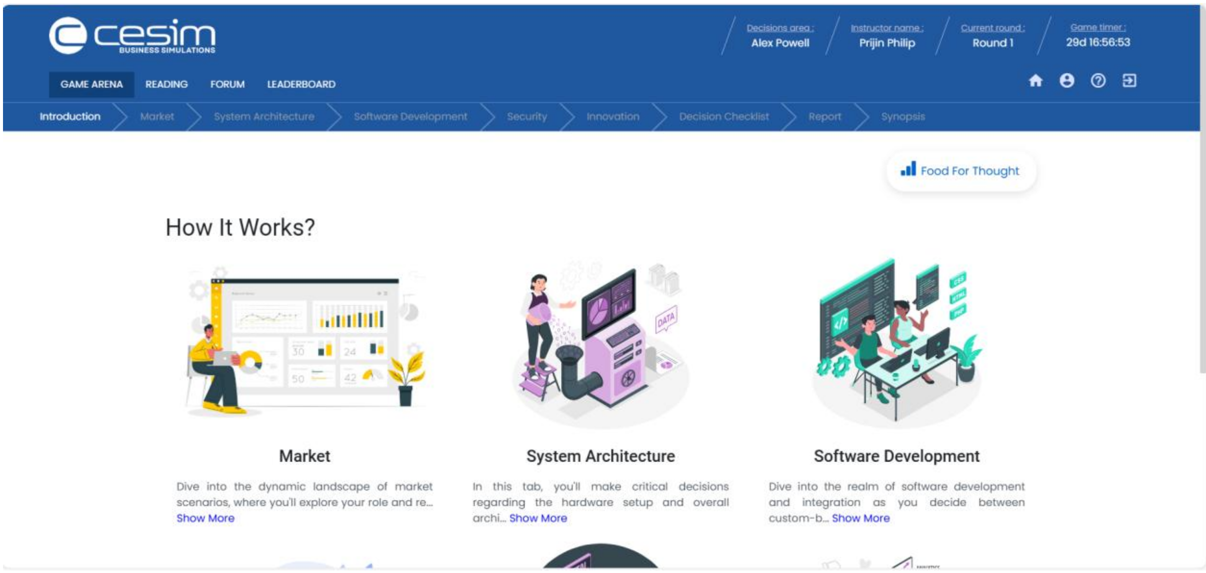


Figure 1 The Game Arena

1.2 Readings

This section provides essential materials necessary for participants to actively engage in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the simulation. This video presents insights into the industry, ongoing trends, obstacles, and the diverse array of decisions required.

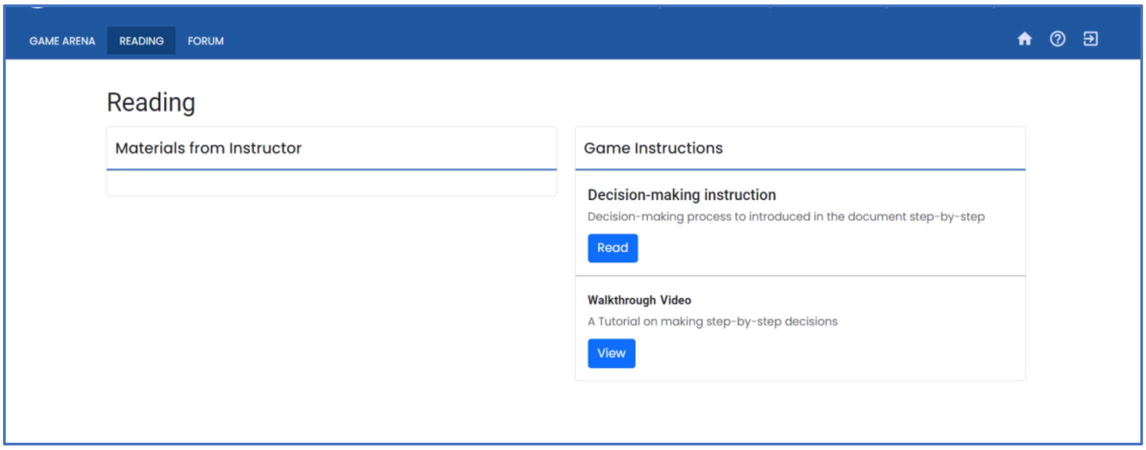


Figure 2 The Reading Tab

1.3 Forum

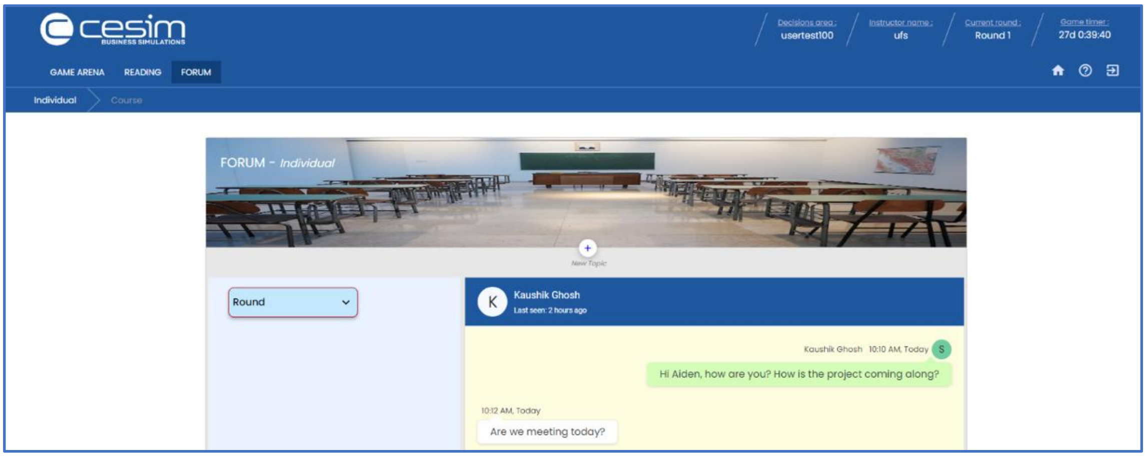


Figure 3 The Forum Tab

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

Silicon Solutions Pvt. Ltd, a leading tech provider in Bangalore, has launched a program to support start-ups with discounted consulting services. PayGuard Technologies Pvt. Ltd., a rising player in the payment gateway industry, has partnered with Silicon Solutions to develop its Management Information System (MIS) with expertise in architecture, database management, security, and compliance. Participants must strategically suggest innovation, security, system architecture and software development that will benefit both the companies.

Success is based on three Key Performance Indicators (KPIs):

1. Performance (%) – Higher is better.
2. Security (%) – Higher is better.
3. Value generation after 3 years.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

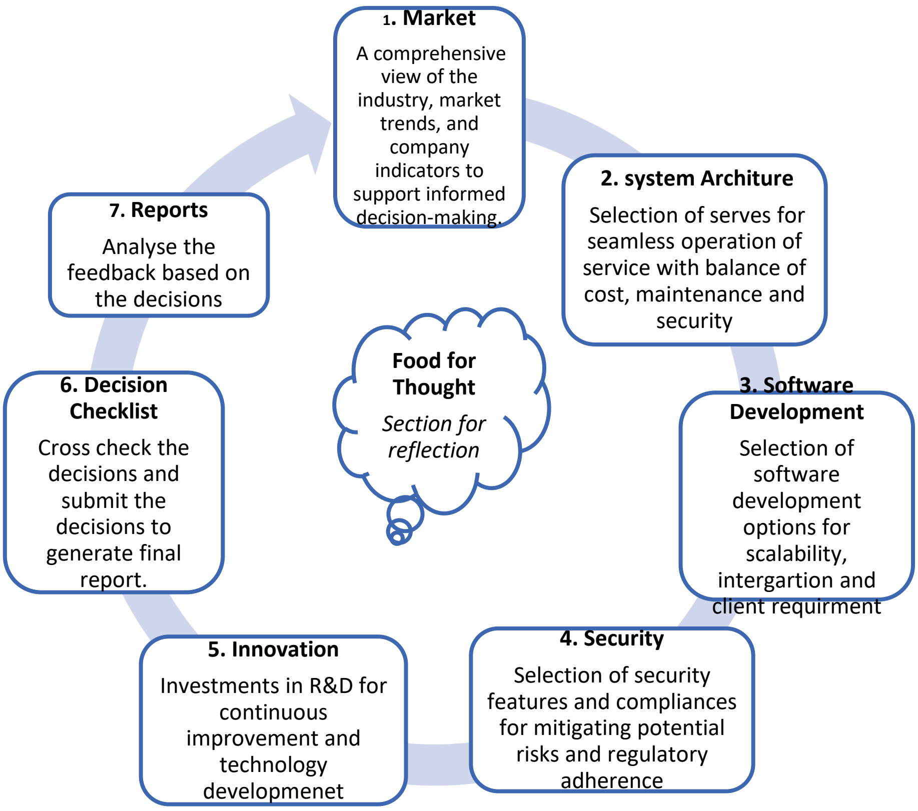


Figure 4 The Sequence of Decision-Making Process

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

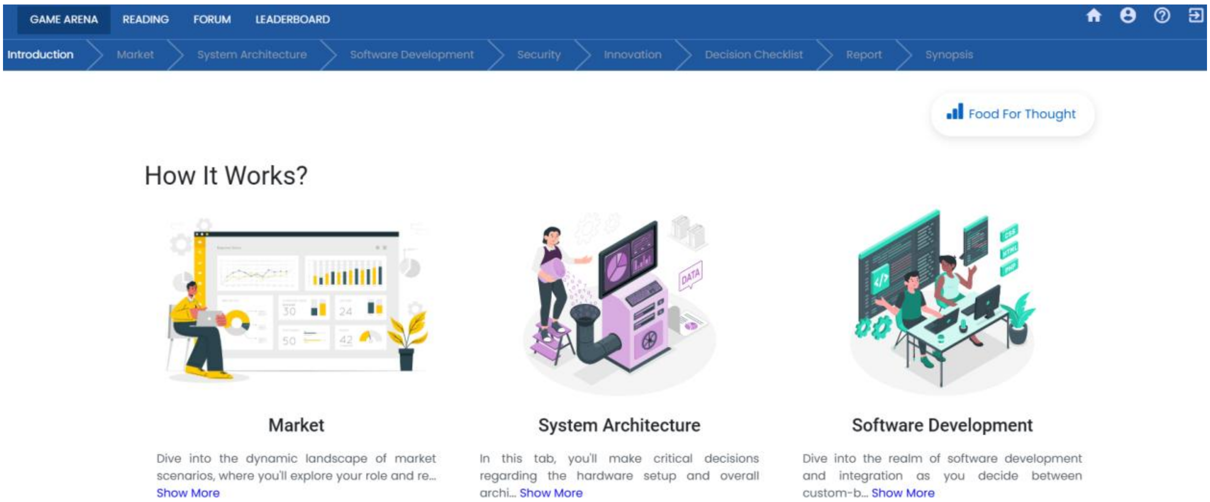


Figure 5 The Introduction Tab in the Game Arena

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making.

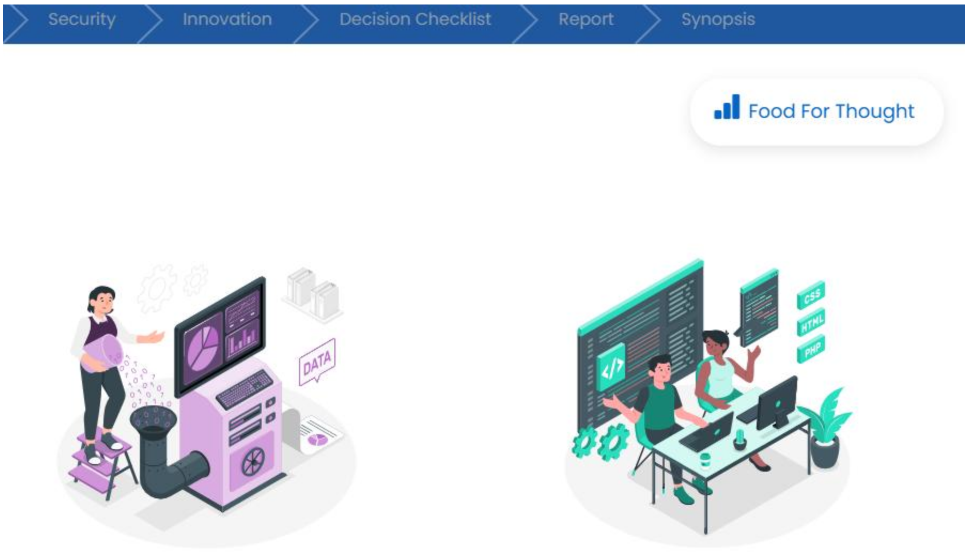
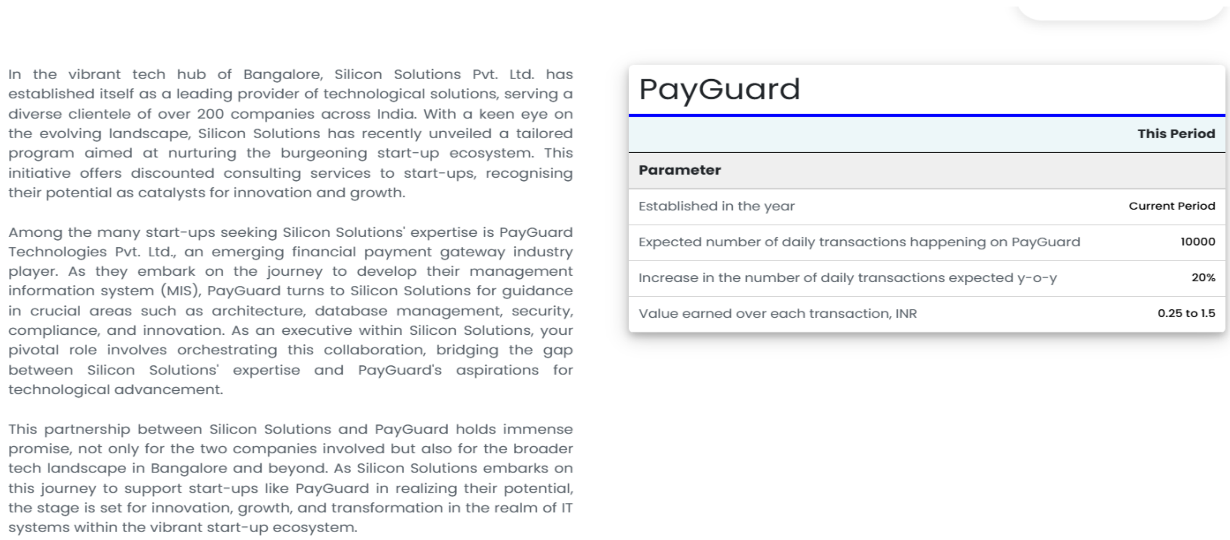


Figure 6 The Food for Thought Section in the Game Arena

2.3 Market

Prior to commencing the decision-making process, it is essential to carefully examine the "Market" section. This section offers vital information on case scenario, and key parameters to consider when making decisions in each tab.



In the vibrant tech hub of Bangalore, Silicon Solutions Pvt. Ltd. has established itself as a leading provider of technological solutions, serving a diverse clientele of over 200 companies across India. With a keen eye on the evolving landscape, Silicon Solutions has recently unveiled a tailored program aimed at nurturing the burgeoning start-up ecosystem. This initiative offers discounted consulting services to start-ups, recognising their potential as catalysts for innovation and growth.

Among the many start-ups seeking Silicon Solutions' expertise is PayGuard Technologies Pvt. Ltd., an emerging financial payment gateway industry player. As they embark on the journey to develop their management information system (MIS), PayGuard turns to Silicon Solutions for guidance in crucial areas such as architecture, database management, security, compliance, and innovation. As an executive within Silicon Solutions, your pivotal role involves orchestrating this collaboration, bridging the gap between Silicon Solutions' expertise and PayGuard's aspirations for technological advancement.

This partnership between Silicon Solutions and PayGuard holds immense promise, not only for the two companies involved but also for the broader tech landscape in Bangalore and beyond. As Silicon Solutions embarks on this journey to support start-ups like PayGuard in realizing their potential, the stage is set for innovation, growth, and transformation in the realm of IT systems within the vibrant start-up ecosystem.

PayGuard

Parameter	This Period
Established in the year	Current Period
Expected number of daily transactions happening on PayGuard	10000
Increase in the number of daily transactions expected y-o-y	20%
Value earned over each transaction, INR	0.25 to 1.5

Figure 7 The Market Tab in the Game Arena

2.4 System Architecture

Under system architecture tab, participants must decide on the best possible options to improve the server for the platform. While deciding on the optimal options, participants should maintain a balance between service scalability, security risk, maintenance cost, performance.

The decision span four key areas:

1. Server Type
2. Server Selection
3. Network Equipment
4. Data Storage & Management

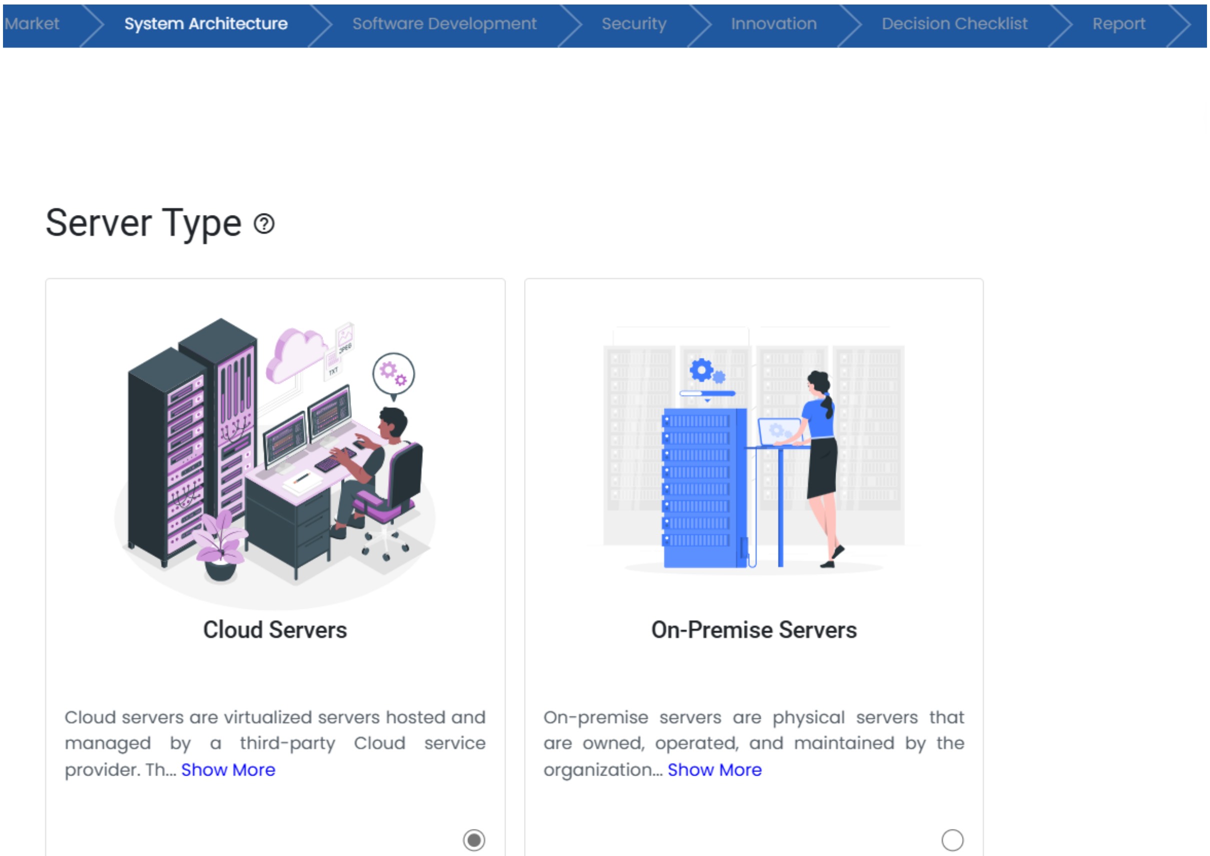


Figure 8 The Decision area under the system architecture tab

2.5 Software Development

Under software development tab, participant will have to take crucial decision on software development of the product. Decision needs to made with long term implications of development models on support, security and margin either off the shelf solution or custom build based on scalability, integration capabilities and customer requirement. Software development decisions will also involve lifecycle management of the project through selecting methods that best suit the business requirements, time constraints, and uncertainty factors.

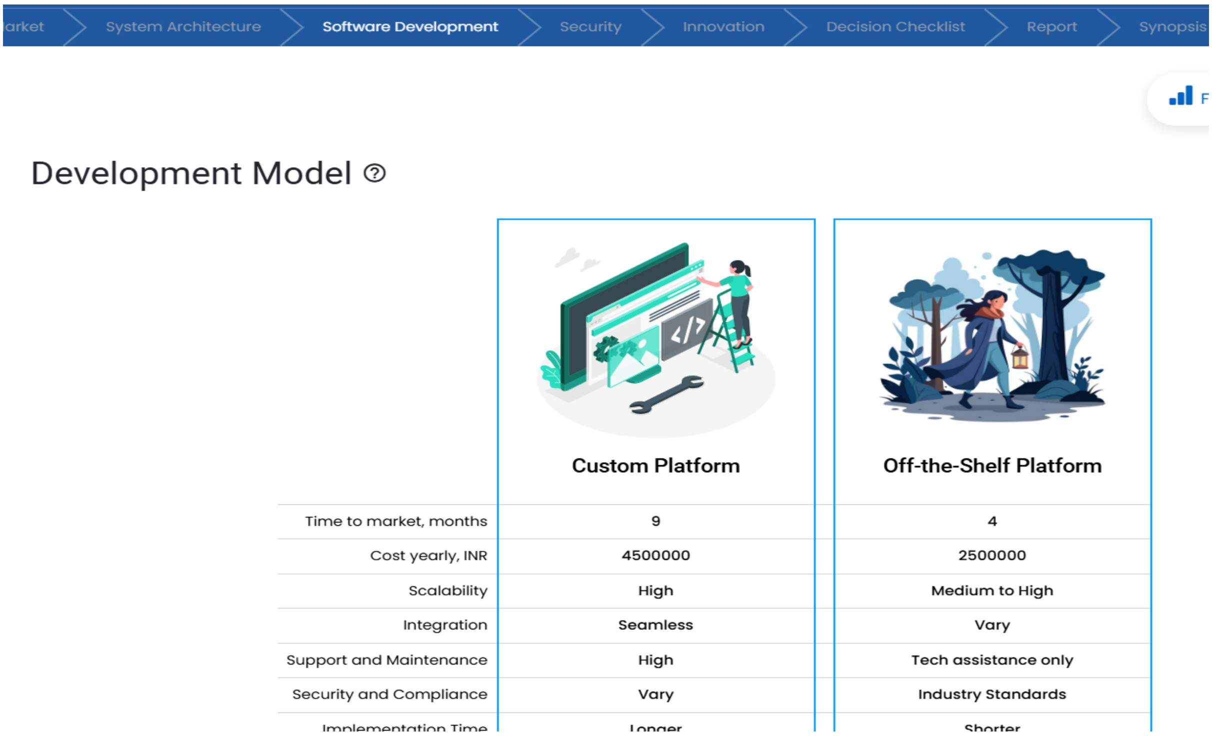


Figure 9 The Decision areas under software development tab

2.6 Security

Participants will have to decide on the security features that align with business requirements, cost considerations, implementation time to mitigate the potential security risks with consideration to system performance and overall business strategy. Regulatory adherence complexity decision should align with cost and time with a measure of impact on development timeline and allocated budget.

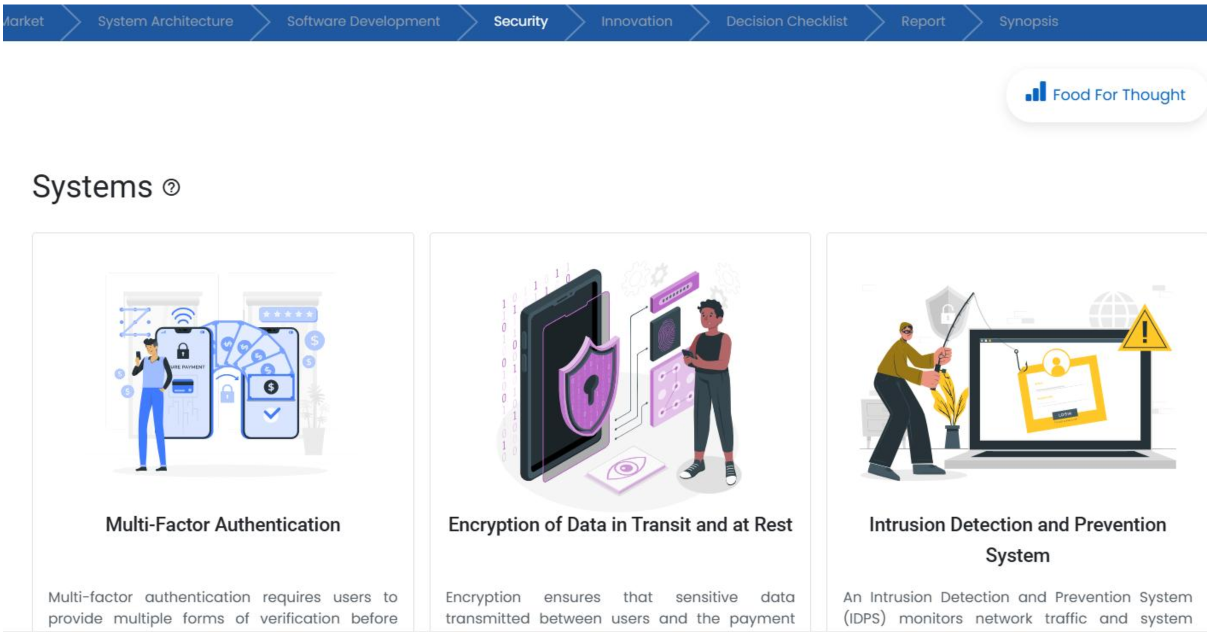


Figure 10 The Decision areas under security tab

2.7 Innovation

Under innovation tab, participants will have to decide on opportunities to innovate new technologies and continue improvement on the existing product. Research and development roadmap decision has not be based on cost, feasibility, development timeframe, technology influence on market opportunities and potential risks. Pilot projects should align with business strategic objective and maximum return of investment. To achieve business objectives, employee effectiveness needs to be enhanced. Effective trainings decisions within budget constraints will help participants to achieve their objectives.

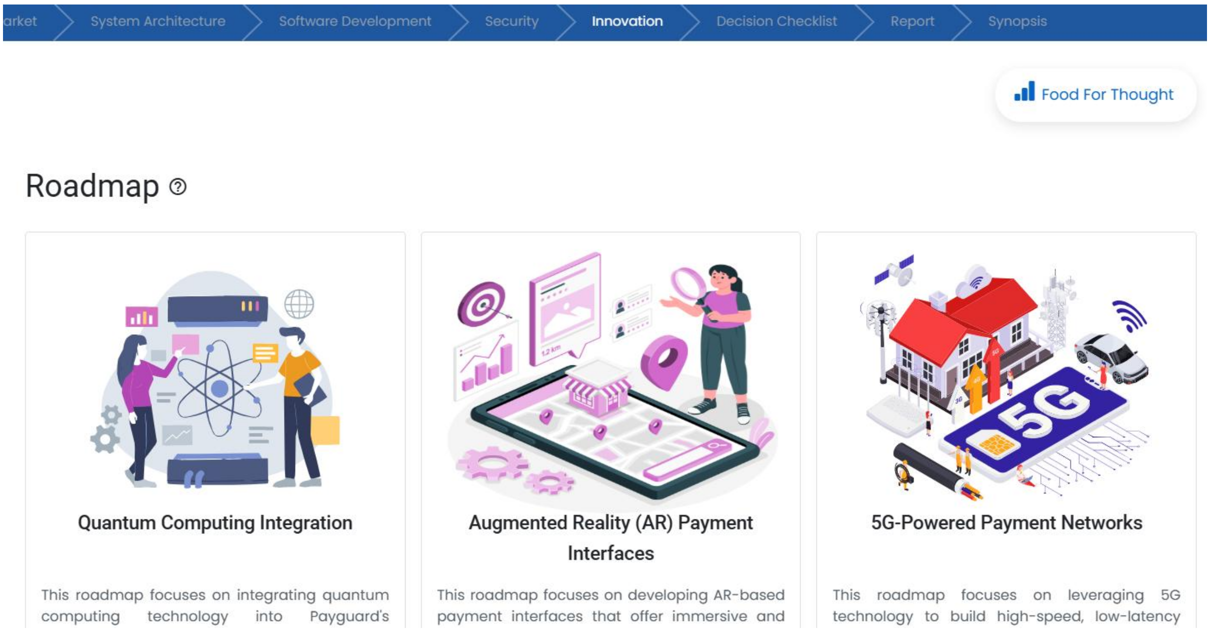
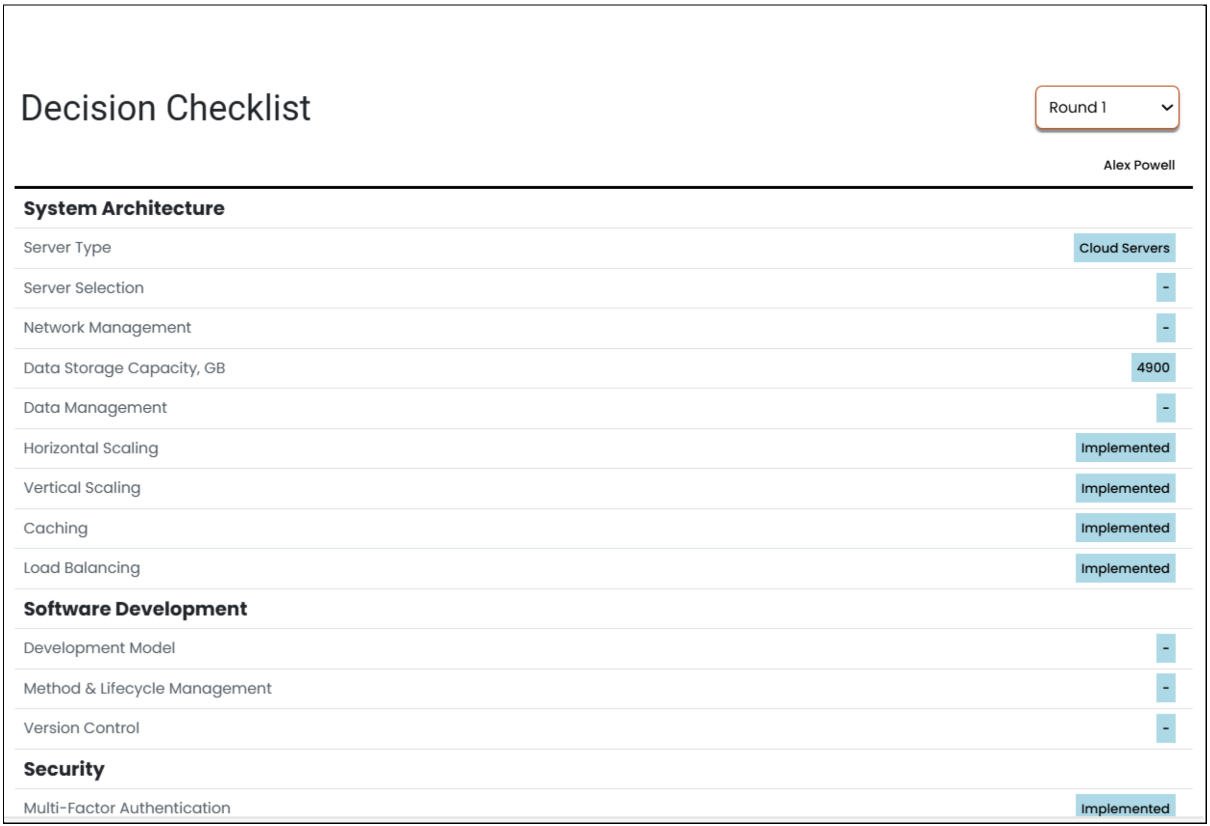


Figure 11 The Decision areas under innovation tab

2.8 Decision checklist

The Decision Checklist assists participants in reviewing and modifying their decisions before the deadline by consolidating them from each decision areas. Once decisions are finalized, participants can submit them using the Submit option if they opt to proceed with the Decision Checklist.



Decision Checklist

Round 1

Alex Powell

System Architecture	
Server Type	Cloud Servers
Server Selection	-
Network Management	-
Data Storage Capacity, GB	4900
Data Management	-
Horizontal Scaling	Implemented
Vertical Scaling	Implemented
Caching	Implemented
Load Balancing	Implemented
Software Development	
Development Model	-
Method & Lifecycle Management	-
Version Control	-
Security	
Multi-Factor Authentication	Implemented

Figure 12 The Decision Checklist

2.9 Report

The report section starts with the participants' critical thinking abilities in the food for thought section based on four metrics.

- Rigour: Consistency throughout the decision-making and recommendation.
- Structuring: Logical approach of gathering and analysing information
- Synthesis: Approach towards interpreting information and making informed decisions
- Business Judgement: Ability to evaluate facts, and consider risks and possible implications when making decisions.

Report

Excellent work. You have completed deciding what the IT infrastructure of the company would look like to make successful deliveries. In this exercise every decision you have taken shows your skills in dimensions like rigor, synthesis, structuring and business judgement.

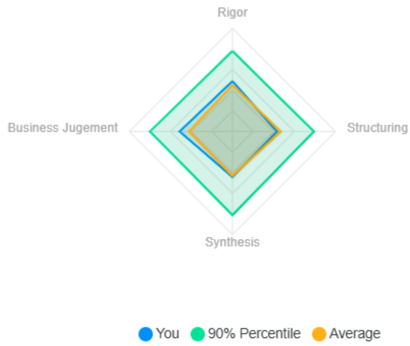
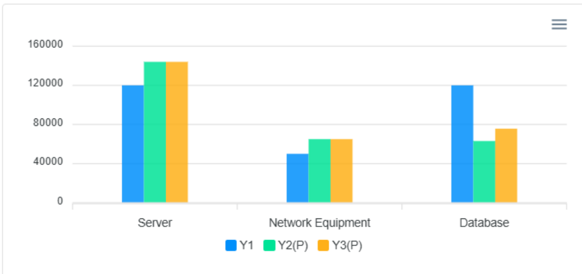


Figure 13 The Report section in the Game Arena

The Report section provides details on the system architecture cost for the current year and planned costs for the next two years. It also outlines the increase in data storage capacity for the current year and future years.

System Architecture Cost, INR

	Y1	Y2 (P)	Y3 (P)
Server	120,000	144,000	144,000
Network Equipment	50,000	65,000	65,000
Database	120,000	63,072	75,686
Total Cost	290,000	272,072	284,686



Data Storage Capacity

	Y1	Y2 (P)	Y3 (P)
Daily Transaction	12,000	14,400	17,280
Required Capacity, GB	4,380	5,256	6,307

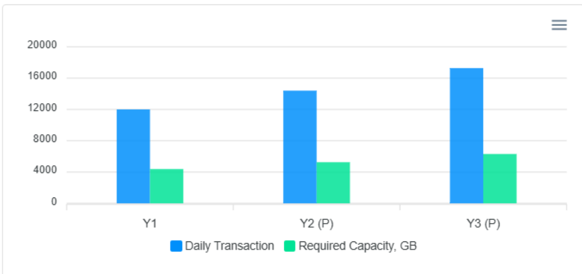


Figure 14 The Report Section in the Game Arena

The Report section also details the costs the company will incur across development and different business areas, along with a three-year projection of value creation through R&D and projects. The value creation projections for the second and third years are based on assumptions that might change with requirements.

Development & Other Cost, INR	
	Y1
Scalability & Performance	60,000
Platform Development	4,500,000
Compliances	600,000
System	700,000
R&D	1,500,000
Pilot Projects	1,300,000
Continous Improvements	300,000
Training & Development	250,000
Total Cost	9,210,000

Projected Value Earned, INR			
	Y1	Y2 (P)	Y3 (P)
Value Earned, INR	6,570,000	7,884,000	9,460,800
Tech Cost, INR	9,500,000	4,890,896	4,571,959

Note : The projections for year 2 and year 3 are based on certain assumptions of business which might change as per requirements.

Figure 15 The Report Section in the Game Arena

The KPI Section indicates the performance metrics encompasses of speed, responsiveness, scalability and user experience. While the security metric cover data encryption, vulnerability management and access control. Ideally both performance and security KPI should be closer to 100% for better outcomes, low percentage and high uptime suggest room for improvement despite perfect uptime score.

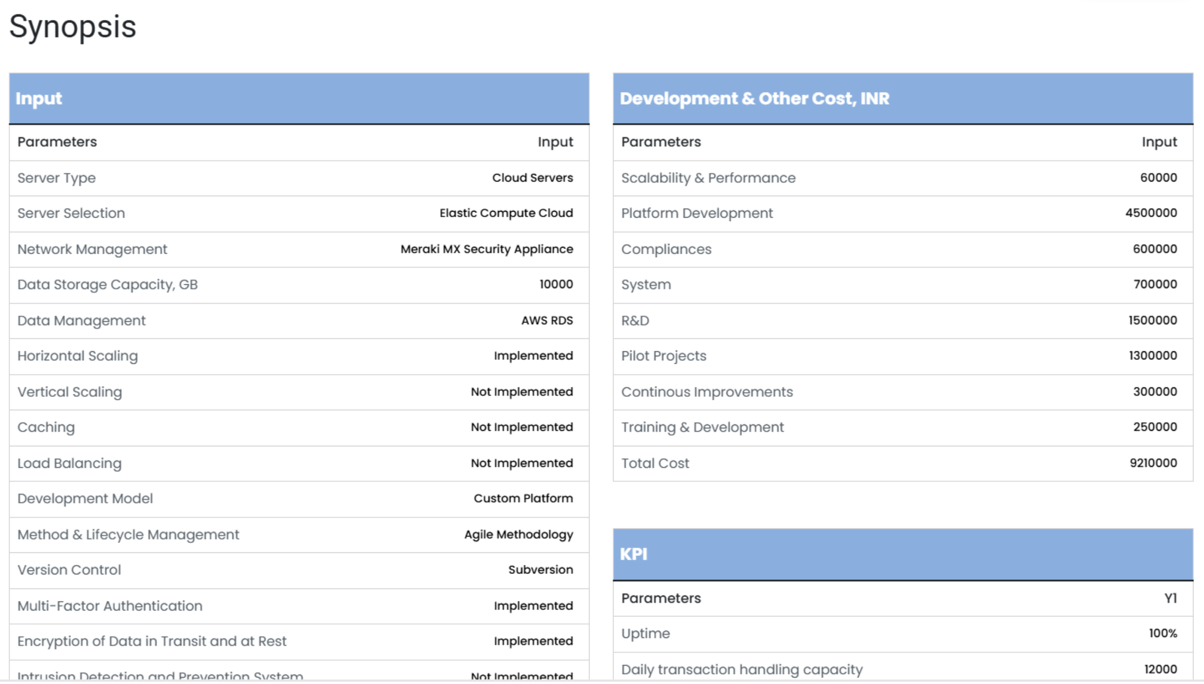
KPI	
	Y1
Uptime	100%
Daily transaction handling capacity	12,000
Compliance timeline, months	6
Minimum development time, months	14
Maximum development time, months	16
Performance	56%
Security	41%

Note : The performance indicates speed, responsiveness, scalability and user experience while the security indicates data encryption, vulnerability management and access control. The closer both KPI's to 100 the better it is from a technology standpoint.

Figure 16 The Report Section in the Game Arena

2.10 Synopsis

The synopsis section provides a summary of the decision areas and the report, highlighting how effectively you applied critical thinking in each aspect, outcomes in different areas, offering insights into the impact of your decision-making skills.



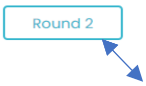
Synopsis

Input	
Parameters	Input
Server Type	Cloud Servers
Server Selection	Elastic Compute Cloud
Network Management	Meraki MX Security Appliance
Data Storage Capacity, GB	10000
Data Management	AWS RDS
Horizontal Scaling	Implemented
Vertical Scaling	Not Implemented
Caching	Not Implemented
Load Balancing	Not Implemented
Development Model	Custom Platform
Method & Lifecycle Management	Agile Methodology
Version Control	Subversion
Multi-Factor Authentication	Implemented
Encryption of Data in Transit and at Rest	Implemented
Intrusion Detection and Prevention System	Not Implemented

Development & Other Cost, INR	
Parameters	Input
Scalability & Performance	60000
Platform Development	4500000
Compliances	600000
System	700000
R&D	1500000
Pilot Projects	1300000
Continous Improvements	300000
Training & Development	250000
Total Cost	9210000

KPI	
Parameters	Y1
Uptime	100%
Daily transaction handling capacity	12000

Figure 17 The synopsis tab



To play the next iteration, follow these steps:

1. Go to the end of the synopsis page, to find the next round.
2. Click on the next round.
3. The participant will be redirected to the home page.
4. Click on the card to start the next round.

Note: The leaderboard for a round will only be updated only once after moving to the next round.